

- It uses the MapReduce programming model to process large data sets.
- Hadoop splits different files into large blocks and then distributes them across various nodes in a cluster.
- It transfers packaged code into different nodes to process the data in parallel.
- Apache Hadoop makes use of the data locality approach, where nodes manipulate all the data they have access to. This allows the large dataset to be processed faster and even more efficiently.
- The base Apache Hadoop framework is composed of different modules: Hadoop Common, HDFS, Hadoop YARN and Hadoop MapReduce.

Cassandra: This is an open source distributed NoSQL database management system. It's designed to handle large amounts of data across many different commodity servers, hence providing high availability with no single point of failure. It offers strong support for clusters that span various data centres, with its asynchronous master less replication allowing low latency operations for all clients.

Features:

- It supports replication and multiple data centre replication.
- It has immense scalability.
- It is fault-tolerant.
- It is decentralised.
- It has tunable consistency.
- It provides MapReduce support.
- It supports Cassandra Query Language (CQL) as an alternative to the Structured Query Language (SQL).

KNIME: Also called Konstanz Information Miner, this is an open source data analytics, integration and reporting platform. It integrates different components for data mining and machine learning through its modular data pipelining concept. A graphical user interface allows the assembly of nodes for data pre-processing (which includes extraction, transformation and loading), data modelling, visualisation and data analysis. Since 2006, it has been widely used in pharmaceutical research, but now it is also used in areas like customer data analysis in CRM, financial data analysis and business intelligence.

Features:

- KNIME is written using Java and is based on Eclipse. It makes use of its extension capability to add plugins, hence providing additional functionality.
- The core version of KNIME includes modules for data integration, data transformation as well as the commonly used methods for data visualisation and analysis.
- It allows users to create data flows and selectively execute some or all of them.
- It allows us to inspect the models, results and interactive views of the flow.
- KNIME workflows can also be used as data sets to create report templates, which can be exported to different document formats like doc, PPT, etc.

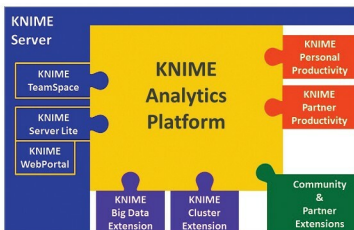


Figure 6: The KNIME analytics platform's architecture
(Image credits: googleimages.com)

- KNIME's core architecture allows the processing of large data volumes which are only limited by the available hard disk space.
- Additional plugins allow the integration of different methods for image mining, text mining as well as time series analysis.

Rapid Miner: This is basically a data science software platform. It is used for business and commercial applications as well as for education, research, rapid prototyping, training and application development. It supports all the steps of the machine learning process including data preparation, model validation, results visualisation and optimisation. It has been developed on an open core model. It provides a graphical user interface to design and execute different analytical workflows. All such workflows are called 'processes' and they consist of multiple 'operators'. Each of these operators perform a single task within their respective processes, and the output of each operator forms the input for the next one. Also, the engine can be called from other programs or can be used as an API.

Features:

- Uses a client or server model with the server offered either on premise, or in private or public cloud infrastructures.
- Ninety-nine per cent of this advanced analytical solution is provided through different template-based frameworks that accelerate delivery and reduce errors by almost eliminating the need to write code.
- Provides various machine learning and data mining procedures including data loading and transformation, predictive analytics and statistical modelling, data pre-processing and visualisation, evaluation and deployment, etc.
- Is written using the Java programming language.
- Provides different learning schemes, algorithms and models, which can be extended using Python and R scripts.
- Its functionality can be easily extended with additional plugins, which are made available via its 'Marketplace', which provides a platform for developers to create data analysis algorithms and then publish them for the community.